Qwen3-VL-8B on Mac mini (MLX): test findings

Date: 2026-10-23 · IDP web extraction (PAN, Aadhaar) · Status: findings only, no decision yet

Setup

Machine	Mac mini "mini3" (10.0.3.66), 24 GB unified memory
Model	Qwen3-VL-8B-Instruct, 4-bit (MLX), 5.4 GB
Engine	MLX `mlx_vlm.server` 0.7.10, continuous batching (`--max-num-seqs 8`)
Test	Same as the earlier tests: 10 real documents (4 PAN, 6 Aadhaar), 20 requests per level, 1 / 2 / 4 / 8 users at the same time, every request a unique copy of its image, same prompts
Image tokens	Same as the Qwen3.5-4B (both cut images into 32 px pieces; e.g. 658 vs 668, 1102 vs 1113), no scaling
Compared with	Qwen3.5-4B and Qwen2.5-VL-3B on MLX, same machine and test (see LLAMACPP_VS_MLX.md, QWEN2.5_VL_3B_MLX.md)

"Typical" = median wait (half the users waited less). "Slow" = 95th percentile (1 in 20 waited longer). The 3B figures are from its matched-token run.

Wait per user

Users at the same time	8B typical / slow	4B MLX typical / slow	3B typical / slow
1	6.3 s / 12.4 s	3.9 s / 7.4 s	3.2 s / 6.3 s
2	10.7 s / 18.4 s	6.1 s / 11.0 s	5.6 s / 9.9 s
4	16.7 s / 34.7 s	9.9 s / 19.7 s	9.6 s / 19.5 s
8	37.3 s / 80.1 s	21.9 s / 40.6 s	18.9 s / 36.2 s

Throughput, accuracy and errors

Users at the same time	8B docs/min	4B docs/min	3B docs/min	8B accuracy	4B accuracy	3B accuracy
1	8.2	13.0	15.7	98.2%	100%	87.5%
2	10.1	16.7	19.1	96.4%	99.1%	84.8%
4	11.1	18.8	20.2	96.4%	99.1%	84.8%
8	10.9	19.1	18.8	95.5%	99.1%	85.0%

Failed requests: none of 80 (every answer was valid JSON).

Accuracy = fields matching the correct values (56 fields per round of the 10 images).

Where the 8B missed

Field	Misses (all 4 levels, 80 requests)
Address (Aadhaar)	14
Gender (Aadhaar)	1 (at 8 users)
All other fields	0

- The address misses were on the same two Aadhaar images (aadhaar_3, aadhaar_4) every time: aadhaar_4 at every level, aadhaar_3 from 2 users up. In the smoke test the 8B returned no address at all for these two; the 4B read both.
- The other 4 Aadhaar addresses matched at 97-100% similarity.
- PAN was read correctly every time.

Observations

- The 8B was the slowest of the three at every level: about 1.6x the 4B's wait (6.3 vs 3.9 s for one user, 37.3 vs 21.9 s at 8 users).
- Throughput stopped rising after 4 users (about 11 documents per minute), against about 19-20 for the 4B and the 3B.
- Accuracy was 95-98%: higher than the 3B (about 85%), a little lower than the 4B (99-100%).
- Almost all misses were one field (address) on two images, so a prompt change for those layouts might change the result.
- No invalid answers, even without schema enforcement (MLX runs with `SEND_SCHEMA=false`).

Limits

- Only PAN and Aadhaar were tested (10 images). Driving licence is not tested yet.
- One machine: Mac mini .66. Other machines will give different numbers.
- Prompts were written and tuned on the Qwen3.5-4B.